

Appendix IV

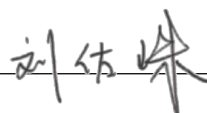


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Literature Review & Project Proposal for ST 4001 Senior Thesis

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Table of Contents

Literature Review

1	Introduction	2
2	Current Research Fields.....	3
2.1	Research direction and current progress	3
2.2	Existing problems and/or obstacles	9
3	Prospectives in the Research Field.....	11

Project Proposal

1	Research Background	12
1.1	Introduction to the research fields	12
1.2	Aim and significance of the proposal.....	13
2	Proposed Research Approaches.....	14
2.1	Major research approaches	14
2.2	Techniques and methodology	14
2.3	Feasibility analysis.....	15
3	Project Schedule and Expected Outcome	16
3.1	Research schedule	16
3.2	Expected outcome	17
4	Research Progress	17
4.1	Preliminary results	17

References

Literature Review

1. Introduction

In recent years, with the advancement in large language models (LLMs), medical artificial intelligence has shift from systems centered on static models to medical AI agents. Unlike conventional medical AI focusing on single-turn question-answering[1], medical AI agents are autonomous and adaptive, able to understand the environment, continuously plan multi-step tasks, and dynamically call external tools during execution[2, 3]. This supports more complex diagnostic and treatment processes and patient management tasks, such as clinical decision support, chronic disease follow-up, medical image analysis, and multi-modal diagnosis.

However, as LLM-driven systems evolve to long-term interaction and continuous decision-making, the inadequate design of memory mechanisms has become a bottleneck restricting their long-horizon reasoning and personalized adaptability[4, 5]. Recent LLM systems primarily rely on limited context windows for immediate inference, making it difficult to effectively store, retrieve, and update patient information, historical interactions, and intermediate reasoning states across different time scales[6, 7]. This context-driven architecture is particularly limited in medical scenarios, as clinical decisions often depend on long-term disease progression, phased examination results, and continuous modeling of individual patient differences.

In the general domain, memory mechanisms are gradually being considered a key component supporting the long-term behavioral consistency and contextual coherence of AI agents. The community has proposed various memory structures for general-purpose AI agents, including retrieval-based memory using external storage[8, 9], structured memory with self-organizing capabilities[10], and agent-centric memory frameworks that can dynamically evolve with interaction[11–13]. While these methods have achieved some success in general tasks and open-domain dialogue, their design goals and evaluation scenarios do not adequately address the long-term disease modeling, individual patient difference representation, and clinical safety

constraints unique to the medical field.

To address these gaps, we propose a memory-augmented agent system for medical reasoning. This system uses LLM as the central hub for reasoning and decision-making. By explicitly modeling and managing medical-related memories across different time scales, the AI agent can continuously accumulate, update, and utilize patient information and medical knowledge during long-term interactions, thereby supporting more stable, coherent, and personalized medical reasoning[7].

Based on this, by reviewing relevant work, we systematically summarize the following three research threads:

1. The overall architecture and capabilities evolution of LLM-driven medical AI agents;
2. Memory modeling methods for LLM-driven AI agents;
3. The practices and limitations of existing medical intelligent agent systems in EHR-centered scenarios.

2. Current Research Fields

2.1 Research direction and current progress

2.1.1 LLM-Driven Medical AI Agent

In the medical field, conversational LLM-driven systems initially focused on weakly interactive tasks such as medical question answering, clinical text understanding, and report generation[14]. These models typically generate responses based on static input, lacking explicit modeling of task states and struggling to drive decision-making in complex clinical workflows[2]. As LLM capabilities improved, researchers realized that relying solely on prompt engineering or fixed reasoning templates was insufficient to support the multi-step decision-making, iterative infor-

mation verification, and continuous interaction with external systems required in real-world medical scenarios[6].

The difference between LLM-based medical agents and standard medical large language models lies in their ability to not only process isolated text inputs but also perceive their surrounding environment and take goal-oriented actions, integrating external knowledge retrieval, planning, and tool usage into their decision-making loop[6]. In the general domain, a recent review defined a conceptual framework comprising four core components: planning (cognitive decision-making), action (executing tasks through tools/interfaces), reflection (perceiving and interpreting multimodal data), and memory (storing and retrieving contextual information)[7]. By integrating all four components, the agent can maintain context during interactions, learn from experience, and adapt to constantly changing clinical scenarios, enabling it to think and strategize rather than follow rigid scripts.

Recent medical agent architectures typically decompose complex tasks into structured subtasks and support dynamic tool usage. For example, an agent might break down a diagnostic process into a series of steps (data collection, hypothesis generation, testing, treatment planning), call upon specialized medical tools at each step, and implemented as a sequential task chain[15]. This clinical planning approach improves error isolation and efficiency by handling each subtask with the appropriate tool or module.

In more complex situations requiring multidisciplinary input, the framework employs a collaborative multi-agent paradigm: different agents assume roles such as radiologists, pathologists, and pharmacists, then communicate their findings to each other to arrive at a comprehensive diagnosis[16, 17]. This multi-agent architecture resembles real-world hospital teams and has been shown to reduce uncertainty by combining expertise from multiple domains. These agents employ advanced reasoning strategies, such as Chain-of-Thought[18] and the ReAct[19] paradigm, to handle clinical decision-making.

Meanwhile, in the general domain, a series of SOTA models with self-search and tool-calling capabilities have demonstrated a significant paradigm shift in agent architecture: models no

longer rely on pre-planned research processes, but can autonomously explore the problem space and gradually form solution strategies through searching, experimentation, and feedback in open environments[20, 21]. The community has shifted focus from how to optimize a fixed reasoning workflow for the model, to how to provide the model with a suitable environment so that it can spontaneously plan and reason effectively.

To support the autonomy of these agents, the general AI agent architecture is tightly integrated with the environment and external tools. External capability interfaces allow the agent to perceive and interact with real-world clinical data systems. For example, the agent can read a patient's electronic health record (EHR) to retrieve medical history and laboratory results[22], use optical character recognition (OCR) technology on scanned reports, or analyze images using visual language models[16]. Through knowledge integration components, the agent can query medical knowledge bases, drug databases, or clinical guidelines in real time and cross-validate its reasoning against credible sources[23]. Simultaneously, action interfaces allow the agent to perform tasks by calling hospital software tools, such as entering medical orders in the EHR[24], scheduling examinations[24], or writing referral letters[25]. By placing the LLM within a rich clinical database and tool environment, the agent can go beyond text-based question answering.

In summary, LLM-based agent architectures in healthcare are evolving towards modular, environmentally embedded designs. This general architecture is the foundation for recent progress in creating AI doctors that collaborate with human clinicians and handle multifaceted clinical tasks.

2.1.2 Agent Memory System

As medical AI agents evolve to long-term interaction and continuous decision-making, memory mechanisms have gradually transformed into core components of intelligent agent systems. General LLMs are limited by fixed context windows, making it difficult to maintain information consistency across different time scales. This is particularly prominent in medical scenarios,

where clinical reasoning heavily relies on the patient’s long-term medical history, periodic examination results, and historical decision-making context[6].

Early solutions rely on Retrieval-Augmented Generation (RAG) or long context window models. However, RAG often only performs semantic matching, lacking an understanding of the temporal dimension[26]; while long context window models, although capable of accommodating more information, are susceptible to the lost-in-the-middle effect[27], leading to the loss of critical diagnostic clues and hallucinations. More importantly, these methods lack the ability of memory evolution, which is the mechanism to modify or update existing knowledge as new evidence (such as new laboratory test results) emerges.

To mitigate context length limitations, conventional memory augmentation methods primarily revolve around context management and retrieval-based memory. For example, works like Context-Folding[28] and ReadAgent[29] utilize reinforcement learning or prompt engineering to encourage the model to fold and summarize long-term interaction content, allowing the agent to retain the most critical information from a long-term conversation within a limited context window; SCM[10] and MemoryBank[8] store conversation history in external vector databases, and reintroduce relevant content into the context through similarity retrieval during the inference stage. These methods improve the long-term consistency of the intelligent agent to some extent, but their memory content organization, update strategies, and usage logic are usually static, and the agent itself has limited autonomy in memory management.

Recent research has begun to propose the concept of Agent Memory, emphasizing that memory should not only serve as a passive storage or retrieval tool, but should become a cognitive resource actively generated, organized, and evolved by the AI agent itself[7]. Firstly, the AI agent needs to determine which information is worth retaining as memory, continuously generating new memory entries based on task requirements and reasoning results, and actively establishing connections with existing memories. Secondly, memory is not a static archive, but dynamically evolves with the accumulation of the agent’s experience, for example, being updated, corrected, or even forgotten by new clues in the conversation. Finally, the intelligent agent needs to evaluate the context during interaction and rationally utilize memory content

to enhance LLM tasks. For example, MemGPT[11] treats memory management as an internal operation executable by the agent, maintaining a task-oriented contextual knowledge base to support longer-term reasoning processes; A-MEM[12] and Mem0[13] explicitly model memory units as knowledge nodes with semantic labels and associative relationships, enabling the agent to actively establish links with existing memories thus achieving structured retrieval; MIRIX[30] and MemoryOS[9] design modular memory architectures for multi-type and multi-modal data, combined with a multi-agent framework, to achieve dynamic memory updating and retrieval processes. Similar designs enable the AI agent to simulate human cognitive characteristics: forming a structured long-term knowledge network through continuous accumulation of experience, rather than simply recalling past conversation summaries.

In medical AI agents, Agent Memory can support continuous modeling of patient status, allowing the AI agent to explicitly utilize past medical history and decision-making context in subsequent reasoning, rather than treating each interaction as an independent task. Specifically, Agent Memory encompasses short-term working memory and long-term episodic memory: short-term working memory includes a task-state cache during reasoning, used to support multi-step reasoning, tool calls, and intermediate conclusion maintenance in the current task; long-term episodic memory is used to store important medical events and knowledge across sessions and time scales, resembling a long-term representation of the patient trajectory[7]. The collaboration of the two enables the intelligent agent to perform detailed reasoning in the current context while maintaining consistent reasoning and personalized decision support over the long term.

Overall, the concept of Agent Memory treats memory as one of the core capabilities of an intelligent agent, rather than a simple engineering patch. Related research lays the theoretical and systemic foundation for building medical intelligent agents that can serve clinical scenarios over the long term.

2.1.3 Medical Agents in EHR-Centered Scenarios

In medical scenarios, Electronic Health Records (EHRs) are the core information carriers connecting patient status, clinical decisions, and medical processes. EHRs typically contain structured data (such as diagnostic codes, laboratory test results, and medication records) and a large amount of unstructured text (such as progress notes, discharge summaries, and imaging reports). It features long time spans, strong information heterogeneity, and significant context dependence[31].

In recent years, research has begun to explore the introduction of LLM-driven agents into EHR-centered medical tasks. Compared to traditional systems based on static models, these medical agents can not only parse complex clinical text, but also progressively integrate patient historical information through multi-round interactions, providing support for diagnosis, treatment decisions, or follow-up management[6]. For example, AgentHospital[23] and DiagGym[24] have built a virtual hospital environment including medical records, treatment processes, and multi-role interactions to evaluate the capabilities of AI agents in a simulated clinical environment; while HealthcareAgent[32] demonstrates the feasibility of integrating patient information through multi-round consultations to improve diagnostic quality. These studies indicate that relying solely on language understanding capabilities is insufficient to support complex medical tasks; agents must possess the ability to continuously model the patient’s historical state.

In EHR-driven scenarios, medical agents usually need to simultaneously handle cross-time event sequences and multi-source information. Without an effective memory mechanism, agents can often only perform local reasoning based on current input, making it difficult to form a coherent understanding of the patient’s overall disease progression. Therefore, EHR scenarios naturally expose the shortcomings of traditional context-driven LLM systems in long-term reasoning[6, 14]. To mitigate this problem, some studies have introduced external memory or retrieval mechanisms to dynamically retrieve historical EHR fragments during the reasoning process. For example, EHRAgent[33] is designed as a tool-using agent, utilizing a symbolic

system to map natural language medical terms to the database schema, thereby solving complex clinical queries; RadAgents[16] integrates multimodal retrieval-augmented generation (Multimodal RAG) to retrieve similar images from a database of confirmed cases for comparison and verification, thus reducing false positives in visual models. However, these methods usually treat EHRs as static document collections, lacking systematic modeling of the temporal relationships between medical events, and agents still struggle to continuously and adaptively track the patient's state.

Therefore, designing medical intelligent agents centered around Agent Memory offers a more promising solution for EHR scenarios. By explicitly representing patient-related information as evolvable memory units, Agent Memory in EHR scenarios not only serves to store factual information but also fulfills the functions of patient state modeling and decision context maintenance. The AI agent can update the patient's state after each interaction or treatment decision and record key changes as long-term event memories, thereby supporting consistent reasoning across sessions.

2.2 Existing problems and/or obstacles

1. **Memory modeling remains at the document level, lacking a unified representation of patient status.**

Recent EHR-driven medical intelligent agent systems external memory modules, reintroducing historical medical records, laboratory results, or clinical notes as retrievable text fragments into the context[6]. As a result, it is difficult for agents to explicitly represent explicit relationships between disease progression, treatment feedback, and adverse events.

2. **Temporal and causal consistency are difficult to guarantee in long-term reasoning.**

Medical decision-making critically depends on temporal ordering and relationships between events, such as identifying which test result triggered a treatment adjustment. However, existing LLM-based agents are prone to hallucination issues in long-term reasoning[26],

including information overwriting, temporal inconsistencies, and erroneous inheritance of outdated states[6], which undermine the consistency and credibility of longitudinal clinical decisions.

3. A significant gap exists between medical AI agents and general Agent Memory modeling.

Although benchmarks such as BRIDGE[34] and MedAgentBench[22] attempt to evaluate medical intelligent agents under complex task settings, most evaluations still focus on single-task or local reasoning accuracy. Consequently, they fail to systematically capture an agent’s ability to continuously model patient states over long-term interactions. This limitation weakens quantitative comparisons between different Agent Memory designs.

4. Safety and interpretability challenges remain unresolved.

In medical scenarios, long-term memory is not merely a performance concern but is directly tied to safety, accountability, and clinical trust. In many existing systems, the provenance of memory contents, the rationale for memory updates, and their influence on downstream decisions are often opaque. Once erroneous or biased information is introduced, its effects may be amplified across subsequent reasoning steps, posing a substantial barrier to real-world clinical deployment[6].

In summary, the core bottleneck of current medical intelligent agents in EHR scenarios does not simply stem from insufficient model capabilities, but rather from the lack of an Agent Memory mechanism that is oriented towards patient status, possesses temporal structure, and has evolutionary characteristics. Our research needs to address how to systematically introduce the concept of general agent memory into EHR-driven medical intelligent agents while ensuring safety and controllability.

3. Prospectives in the Research Field

As LLM-driven medical agents are gradually applied to real clinical workflows, future development will primarily focus on the following aspects:

- Instead of treating EHRs as a collection of searchable text, future systems will need to be patient-centric, structurally modeling clinical events across time to support long-term, coherent medical reasoning.
- The design of Agent memory for medical scenarios will become a key research direction. The memory mechanism needs to support both short-term reasoning and long-term disease course modeling, and possess the ability to dynamically update with new evidence to avoid inconsistencies and error accumulation caused by static memory.
- The evaluation system will also evolve accordingly. Long-term interactive benchmarks centered on EHRs will help comprehensively assess the capabilities of medical agents in long-term reasoning and state maintenance, driving the development of medical agents from short-term auxiliary tools to long-term intelligent decision support systems.

Project Proposal

1. Research Background

1.1 Introduction to the research fields

In recent years, the rapid development of Large Language Models (LLMs) has significantly advanced the application of artificial intelligence (AI) in the medical field. From medical question answering and clinical text understanding to assisted diagnosis and report generation, LLM-based systems have demonstrated performance comparable to or even surpassing traditional methods in various tasks[14]. However, most existing medical AI systems still primarily exist as static models or single-turn reasoning tools, making it difficult to adapt to the long-term, continuous, and highly context-dependent decision-making needs of real clinical environments.

As the research paradigm evolves from models to agents, medical agents are gradually becoming a new research hotspot[6]. Medical AI agents can perform multi-step planning, call external tools, and continuously execute tasks within the environment, thus more closely resembling real medical workflows. Meanwhile, clinical practice heavily relies on the integration and dynamic updating of long-term patient information, making Electronic Health Records (EHRs) the core environment for medical agent operation. However, the long time span, strong information heterogeneity, and significant context dependence of EHRs also pose higher demands on the agent's long-term reasoning and state maintenance capabilities[2].

Current research shows that memory augmented methods relying solely on context windows or similarity-based retrieval are insufficient to support continuous modeling of a patient's disease progression[26, 27]. Agents can often retrieve relevant historical information, but struggle to form a stable, evolvable representation of the patient's state internally, thus affecting the consistency and reliability of long-term reasoning. Therefore, how to introduce memory mechanisms for long-term medical reasoning into medical agents has become a key bottleneck limiting their

application in real clinical settings.

Against this backdrop, Agent Memory, as an emerging research direction, provides a new solution for medical agents. By treating memory as an internal state actively generated, organized, and updated by the agent, rather than a passively retrieved external resource[7], Agent Memory has the potential to support long-term modeling of patient states and cross-session reasoning. Exploring memory-augmented medical agents in the EHR scenario not only has significant research value but also has practical implications for building long-term stable intelligent medical decision support systems.

1.2 Aim and significance of the proposal

This study aims to explore Medical Reasoning with a Memory-Augmented Agent System, focusing on how agent memory can support long-term modeling and continuous reasoning of patient states based on EHRs.

This project helps bridge the gap between general agent memory methods and practical medical needs, providing a more structured and interpretable system design approach for long-term medical reasoning. From an application perspective, medical agents with long-term memory capabilities have potential value in scenarios such as chronic disease management, long-term follow-up, and complex case analysis. The exploration in this study can provide methodological and practical references for the evolution of medical intelligent agents from short-term auxiliary tools to long-term intelligent collaborative systems.

2. Proposed Research Approaches

2.1 Major research approaches

This study adopts a research approach combining system design and experimental verification. The overall idea is to build a memory-augmented medical intelligent agent framework based on existing LLMs with reasoning and tool-calling capabilities, and systematically analyze its role in long-term patient state modeling.

The research will use EHR data as the primary application scenario, building the intelligent agent around Agent Memory, enabling the agent to continuously accumulate, update, and utilize patient-related information during multi-round interactions and multiple decision-making processes. In addition, an automated evaluation system will be designed based on existing EHR datasets to generate long-term, multi-session research tasks aligned with real-world medical scenarios, allowing for a multi-dimensional comprehensive evaluation of the intelligent agent architecture. By comparing different memory strategies (e.g. no memory, retrieval-based memory, and Agent Memory), the study will analyze the impact of memory mechanisms on long-term reasoning consistency and decision quality.

2.2 Techniques and methodology

This study will adopt a modular medical intelligent agent architecture, mainly comprising the following key components:

- **LLM Inference and Decision Module.** An existing general-purpose large language model will be used as the core of the intelligent agent's reasoning, responsible for medical language understanding, diagnostic and treatment decision-making, and action planning. It will fully utilize the model's inherent thinking, planning, and tool-calling capabilities without requiring parameter training of the model itself.

- **EHR Environment.** A structured/semi-structured EHR dataset will be constructed or utilized, providing patient history, examination results, and treatment records to the intelligent agent in a queryable format, enabling it to obtain necessary information through tool calls.
- **Agent Memory Module.** An explicit memory management mechanism will be designed, based on the practical needs of medical-oriented long-term interaction, creating a modular, multi-type, and multi-modal memory architecture. The intelligent agent will decide on memory writing, updating, and retrieval based on task requirements, rather than relying solely on passive retrieval.
- **Benchmark Method.** By constructing multi-round diagnosis and treatment or long-term follow-up tasks, the performance of the intelligent agent under different memory configurations will be compared in terms of information consistency, historical utilization rate, and reasoning coherence. A qualitative and quantitative analysis of the role of Agent Memory in the EHR scenario will be conducted.

2.3 Feasibility analysis

From a technical feasibility perspective, this study does not rely on model training or clinical deployment. It is primarily based on existing LLMs, open-source toolchains, and publicly available medical datasets, making the development cost and implementation difficulty manageable. The design and implementation of Agent Memory can be completed gradually through a modular approach, ensuring good scalability.

From an experimental feasibility and risk perspective, EHR data can be obtained through de-identified public datasets or simulated environments, avoiding the compliance risks associated with real clinical data. Furthermore, evaluation tasks can be completed in offline or simulated environments, making implementation straightforward.

3. Project Schedule and Expected Outcome

3.1 Research schedule

- **Project Initiation** (Oct. 18, 2025 – Nov. 15, 2025): Complete project confirmation and research scope definition, clarifying research objectives and overall technical roadmap. Subsequently, during the literature review phase (Nov. 16, 2025 – Jan. 7, 2026), systematically review the research progress of medical datasets, LLM-based agents, and memory-augmented reasoning frameworks, focusing on analyzing the shortcomings of existing methods in terms of long-term reasoning continuity and interpretability. Based on this, complete the writing and submission of the thesis proposal in mid-January 2026 (Jan. 8, 2026 – Jan. 23, 2026).
- **System Development** (Jan. 24, 2026 – Mar. 20, 2026): Design the overall architecture of the memory-augmented medical intelligent agent, clarifying the module division of Agent Memory, the EHR data interface format, and the agent's operation process; construct multi-round diagnosis or long-term follow-up experimental tasks to conduct comparative experiments on different memory strategies. Complete a preliminary demonstration of the system prototype during the mid-term review stage (Mar. 27, 2026), and make necessary adjustments and optimizations to the research methods and system architecture based on feedback.
- **Experimental Implementation and Results Analysis** (Mar. 28, 2026 – May 10, 2026): Conduct system experiments on benchmark datasets or simulated EHR scenarios, comparing and analyzing the performance of the memory-augmented intelligent agent and baseline LLM methods in terms of diagnostic reasoning accuracy, temporal consistency, and explanation quality, and systematically analyze and summarize the experimental results.
- **Thesis Writing** (May 11, 2026 – May 18, 2026): Systematically organize research findings, complete the first draft of the graduation thesis, and revise and refine it through

multiple rounds of revisions based on the supervisor's feedback, preparing for the final submission and defense of the thesis.

3.2 Expected outcome

- We propose a memory-augmented medical intelligent agent system framework for EHR scenarios, clarifying the role of Agent Memory in long-term medical reasoning.
- We implement a working prototype of the medical intelligent agent system to verify the effectiveness of the memory augmentation mechanism in patient state modeling and long-term reasoning consistency.
- Through experimental comparative analysis, we summarize the advantages and disadvantages of existing medical intelligent agents in long-term reasoning and memory modeling, providing a reference for future research.

4. Research Progress

4.1 Preliminary results

We have reviewed research progress in medical agents, LLM-based agents, and Agent Memory, focusing on the shortcomings of existing methods in long-term reasoning and patient state modeling within EHR scenarios.

We have completed the preliminary architectural design of a memory-augmented medical agent, defining the core system modules, including the LLM reasoning and decision-making module, the EHR data access interface, and an Agent Memory management mechanism designed for the medical environment.

We have surveyed EHR datasets and have preliminarily designed multi-round diagnosis and long-term tasks as evaluation scenarios. This study selects MIMIC-IV[35] as the initial EHR

data sample to support experimental verification of the medical agent in long-term patient information modeling scenarios. For the architectural design, we referenced A-MEM[12] and completed the local deployment, qualitatively analyzing the characteristics of Agent Memory compared to traditional LLMs in general domain. In addition, we tested a series of medical agent demo systems[36, 37] to deepen our understanding of the general design principles and implementation methods of multi-agent architectures in medical scenarios.

Overall, the current research work has completed the problem analysis and preliminarily verified the rationality and feasibility of the research approach. The next stage will focus on advancing the implementation and experimental verification of the system prototype to further evaluate the actual effectiveness of Agent Memory in supporting long-term medical reasoning in EHR scenarios.

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